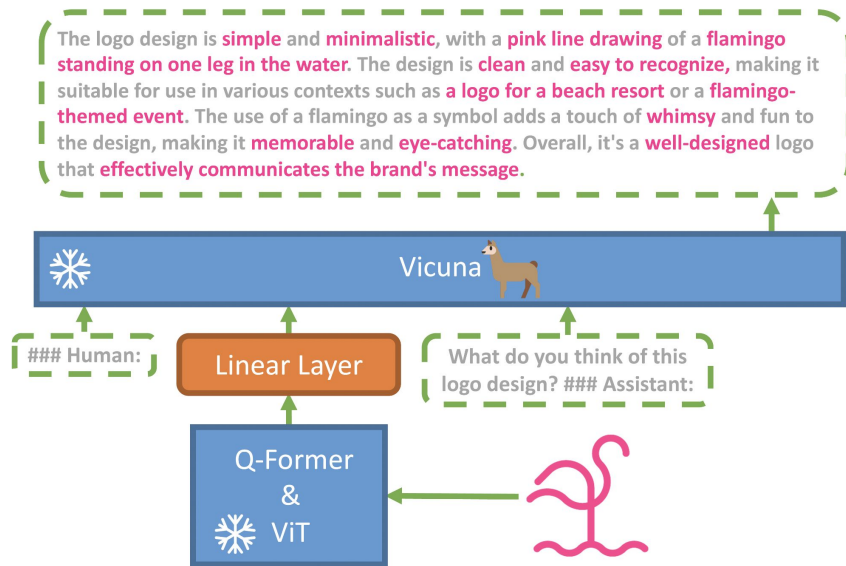


DreamPRM: Domain-Rewighted Process Reward Model for Multimodal Reasoning

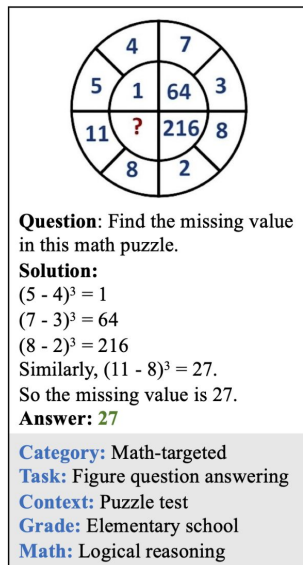
Qi Cao, Ruiyi Wang, Ruiyi Zhang, Sai Ashish Somayajula, Pengtao Xie

Multimodal Language Models (MLLM)

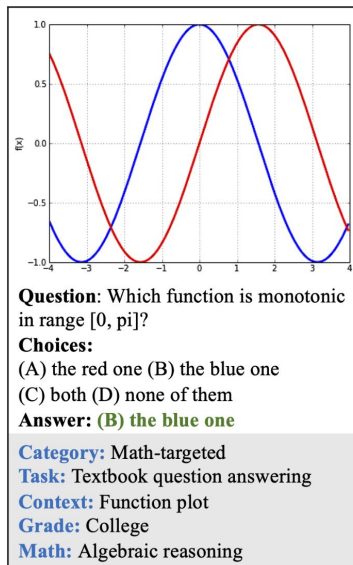


Example of a vision-language model. (Image credit: MiniGPT-4)

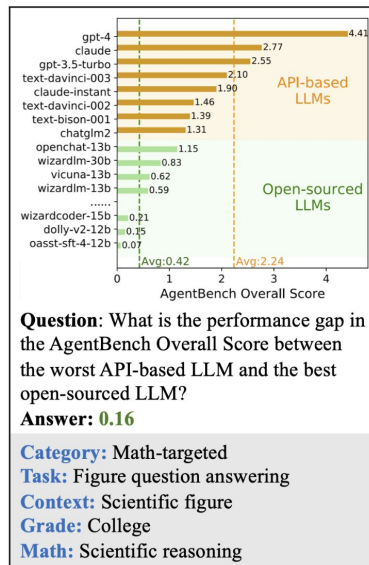
MLLM for Multimodal Reasoning



(a) IQTest



(b) FunctionQA



(c) PaperQA

Examples from MathVista benchmark (Image credit: MathVista)

Chain-of-Thought (CoT) Reasoning

5-step structural thinking for multimodal reasoning

You have been given a question that involves both an image and a text.

Your task is to analyze the question by following exactly five steps:

Step 1: Restate the question.

- Clearly rephrase or clarify the question in your own words.

Step 2: Gather evidence from the image.

- Describe any relevant visual details (e.g., objects, people, locations, interactions) that might address the question.

Step 3: Identify any background knowledge needed.

- Note any general facts, assumptions, or external knowledge that is necessary to address the question.

Step 4: Reason with the current evidence.

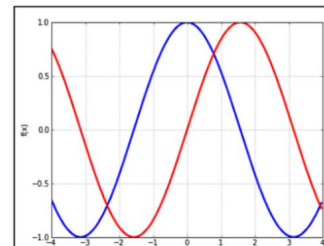
- Integrate the information from the image, text, and relevant background knowledge.
- Show how these pieces of evidence lead toward an answer.

Step 5: Summarize and conclude with all the information.

- Provide a concise, direct answer to the question, referencing the supporting evidence and reasoning.

Once you have completed your reasoning, provide your final answer in the format:

Final answer: ...



Question: Which function is monotonic in range $[0, \pi]$?

Choices:

(A) the red one (B) the blue one
(C) both (D) none of them

Answer: (B) the blue one

Category: Math-targeted

Task: Textbook question answering

Context: Function plot

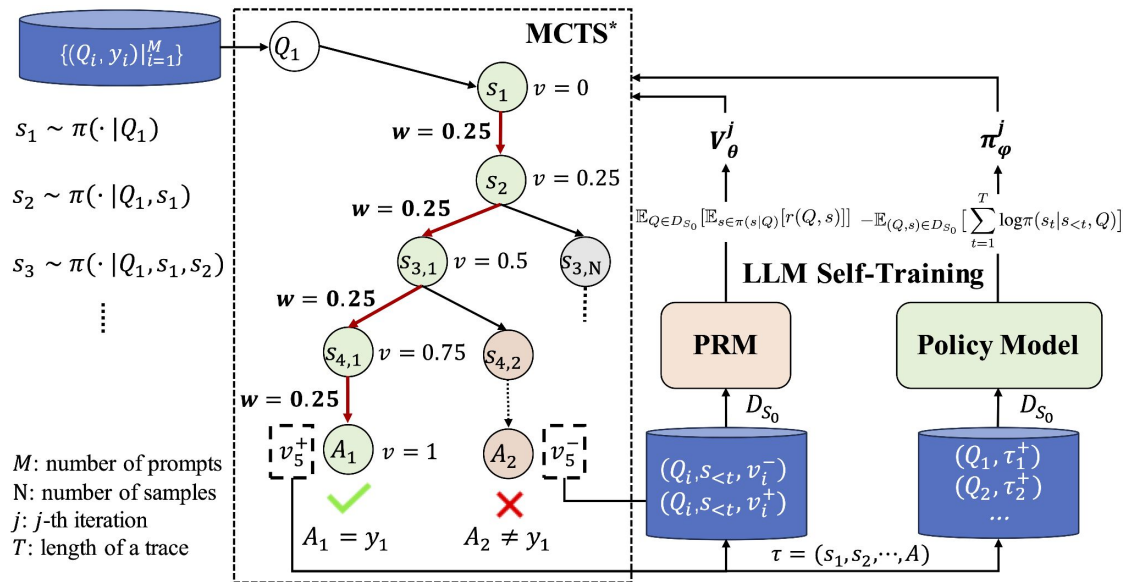
Grade: College

Math: Algebraic reasoning

MathVista

An example of an MLLM prompted to answer a question step by step.

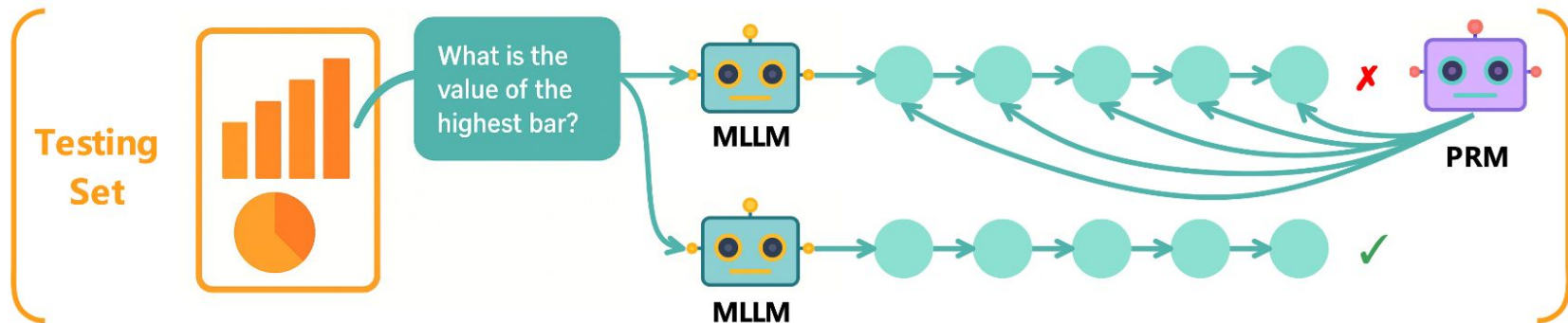
Process Reward Model (PRM)



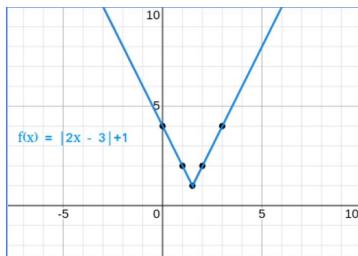
Process Reward Model (PRM) evaluates quality of any partial solution.

Image Credit (ReST-MCTS*)

MLLM+PRM for Reasoning



Use a PRM to rank CoTs generated by an MLLM and select the best CoT.



Question:

The derivative of $f(x)$ at $x=2$ is ____ that at $x=5$

Choices:

(A) larger than (B) equal to (C) smaller than

Ground Truth: B

Benchmark: MathVista

Step 1: Restate the question.

- The question is asking whether the derivative of the function $f(x) = |2x - 3| + 1$ at $x = 2$ is larger than, equal to, or smaller than the derivative at $x = 5$.

[DreamPRM: 0.628]

Step 2: Gather evidence from the image.

- The image shows the graph of the function $f(x) = |2x - 3| + 1$.
- The graph has a V-shape, indicating it is an absolute value function.

[DreamPRM: 0.575]

Step 3: Identify any background knowledge needed.

- The derivative of an absolute value function changes at the point where the expression inside the absolute value is zero.
- For $f(x) = |2x - 3| + 1$, the expression inside the absolute value is $2x - 3$.
- The derivative of $f(x)$ is 2 for $x > 1.5$ and -2 for $x < 1.5$.

[DreamPRM: 0.598]

Step 4: Reason with the current evidence.

- The point where the derivative changes is at $x = 1.5$.
- For $x = 2$, the function is in the interval where $x > 1.5$, so the derivative is 2 .
- For $x = 5$, the function is also in the interval where $x > 1.5$, so the derivative is 2 .

[DreamPRM: 0.748]

Step 5: Summarize and conclude with all the information.

- Both the derivatives at $x = 2$ and $x = 5$ are 2 , so they are equal.

[DreamPRM: 0.812]

Final answer: B ✓

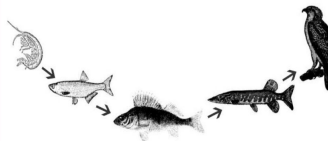
An example of using
a PRM on CoTs

Data Quality Imbalance Problem

Quality imbalance problem in multimodal reasoning datasets:

- Upper: an easy AI2D question.
- Lower: a hard M3CoT question.

Paying too much attention to easy questions can degrade PRM performance.



Question: What does the bird feed on?
Choices:
A. zooplankton B. grass
C. predator fish D. none of the above
Answer: C **Dataset:** AI2D (2016)

Dataset difficulty: easy (InternVL-2.5-MPO-8B's accuracy 84.6%)
Unnecessary modality: can answer without image
Requirements for reasoning: do not require complicated reasoning
Domain weight: 0.55 (Determined by DreamPRM)



Question: Determine the scientific nomenclature of the organism shown in the primary image.
Choices:
A. Hemidactylus turcicus B. Felis silvestris
C. Macropus agilis D. None of the above
Answer: D **Dataset:** M3CoT (2024)

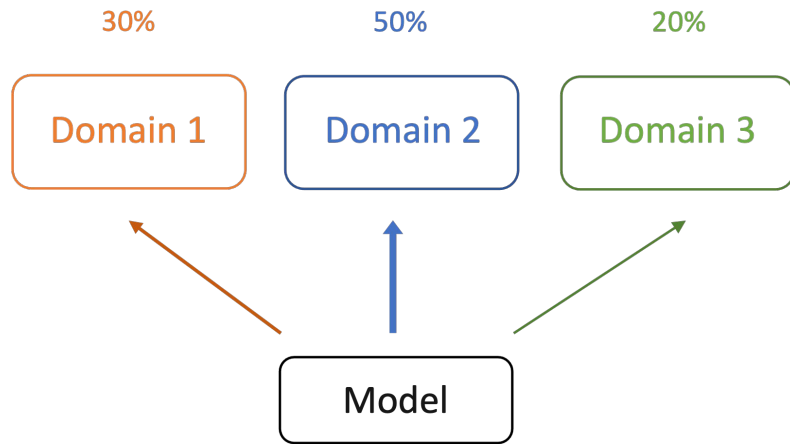
Dataset difficulty: hard (InternVL-2.5-MPO-8B's accuracy 62.1%)
Unnecessary modality: cannot answer without image
Requirements for reasoning ability: require complicated reasoning
Domain weight: 1.49 (Determined by DreamPRM)

Domain Reweighted Training

Assign higher weights to more challenging domains during training.

However, manually assigned weights may be suboptimal.

Idea: Use automatic reweighting.



PRM Training with Monte-Carlo Signals

Problem: Let $p(x)$ be a monic polynomial of degree 4. Three of the roots of $p(x)$ are 1, 2, 3. Find $p(0)+p(4)$.

Golden Answer: 24

Solution: Since three of the roots of $p(x)$ Final Answer 20. ✗

Problem:
Since three of the roots of $p(x)$ are 1, 2 and 3, we can write:

Rollout 1: Final Answer 24. ✓

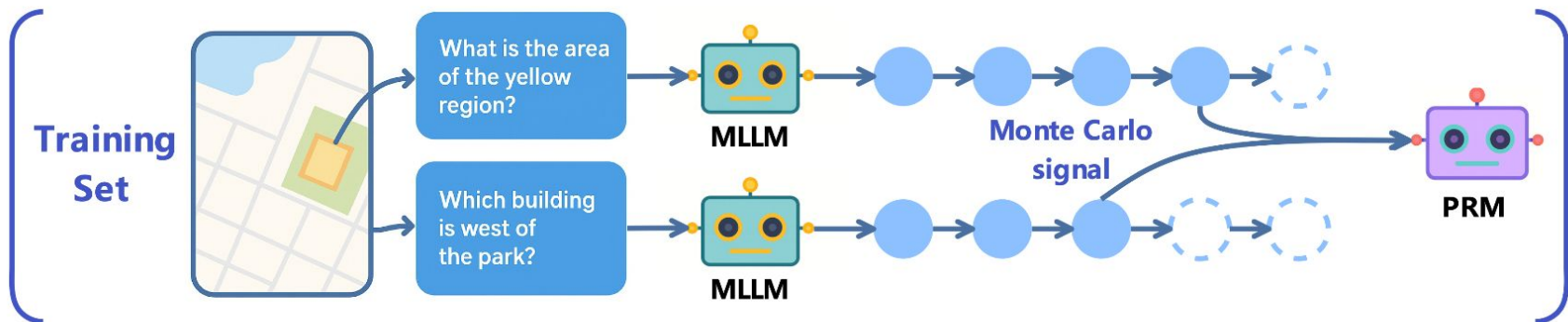
Rollout 2: Final Answer 24. ✓

Rollout 3: Final Answer 20. ✗

$MC = 0.67$

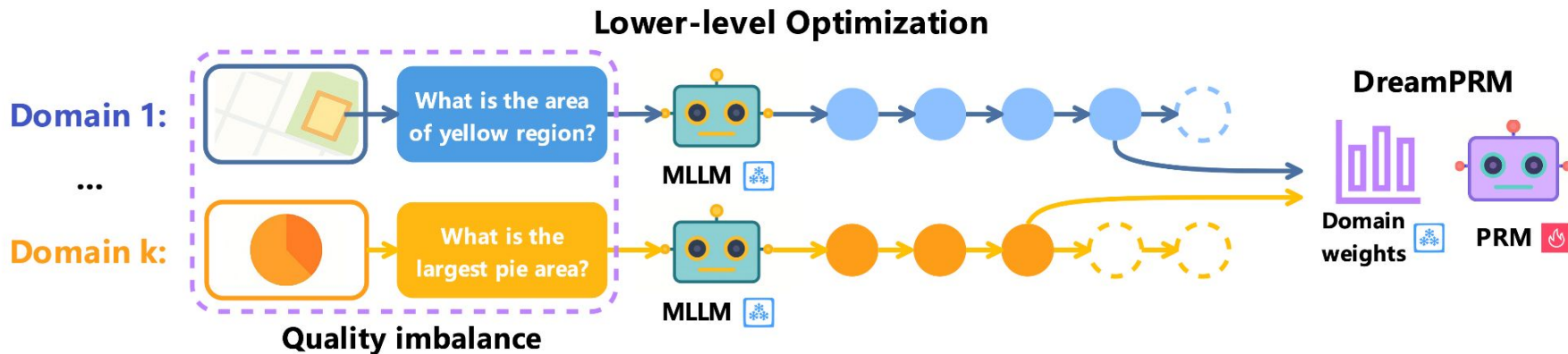
(Image Credit: Google DeepMind)

PRM Training with Monte-Carlo Signals (2)



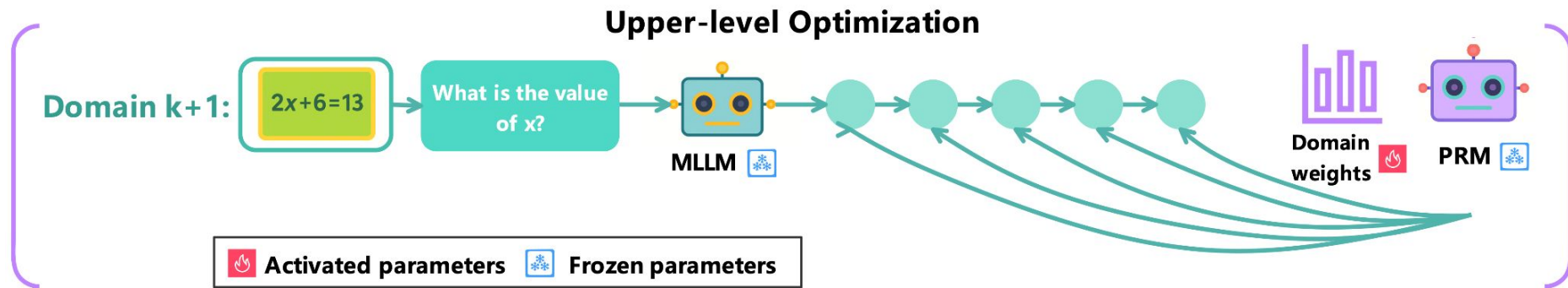
Use Monte-Carto signals as labels to train PRM.

DreamPRM: Lower-level Optimization



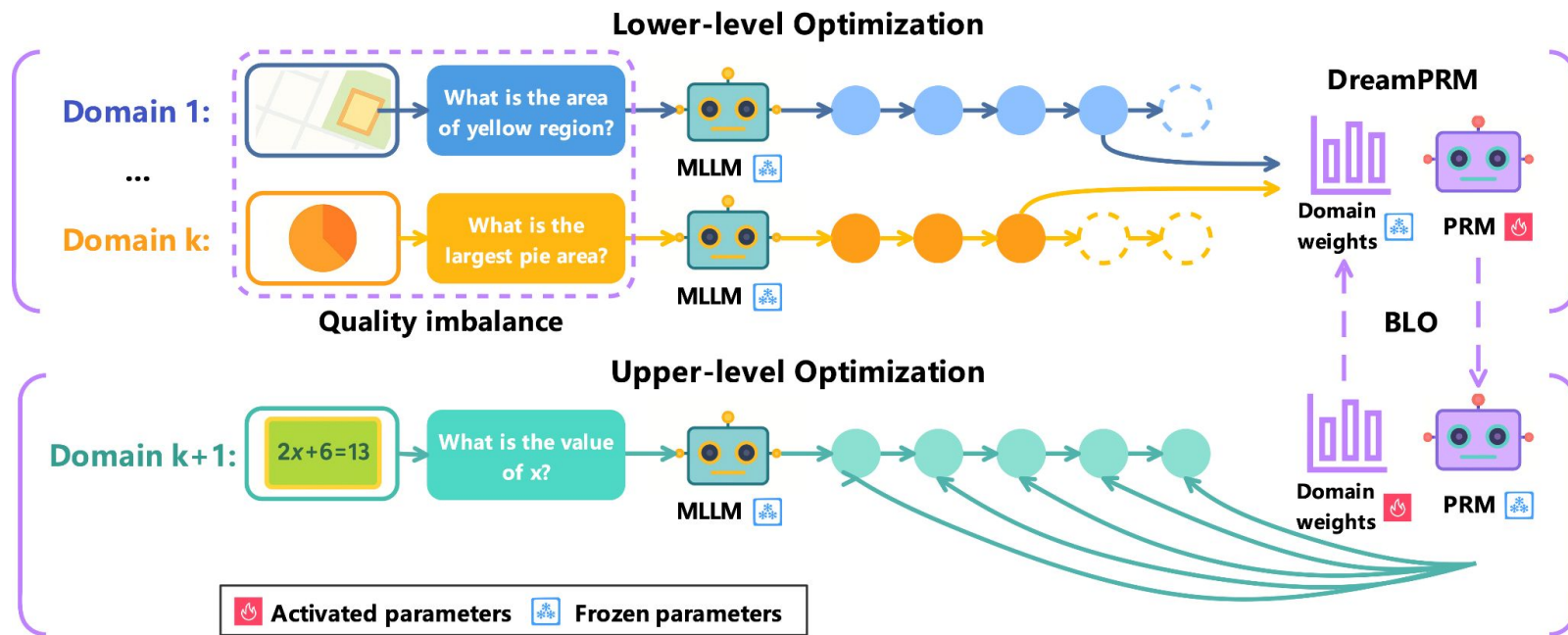
Optimize PRM parameters with domain reweighted training.

DreamPRM: Upper-level Optimization



Optimize domain weights on a separate meta domain.

DreamPRM: Bi-level Optimization



DreamPRM: Bi-level Optimization (2)

$$\begin{aligned} & \min_{\alpha} \mathcal{L}_{meta}(\mathcal{D}_{meta}, \phi^*(\alpha)) \\ s.t \quad & \phi^*(\alpha) = \arg \min_{\phi} \mathcal{L}_{tr}(\mathcal{D}_{tr}, \phi, \alpha) \end{aligned}$$

α : trainable domain weights

ϕ : PRM parameters

DreamPRM Ranks First in MathVista Leaderboard

Leaderboard on MathVista (testmini)

Accuracy scores on the **testmini** subset (1,000 examples) of  MATHVISTA.

#	Model	Method	Source	Date	ALL	FQA	GPS	MWP	TQA	VQA	ALG	ARI	GEO	LOG	NUM	SCI	STA
-	Human Performance*	-	Link	2023-10-03	60.3	59.7	48.4	73.0	63.2	55.9	50.9	59.2	51.4	40.7	53.8	64.9	63.9
1	DreamPRM (o4-mini) 🏆	Reason 🧠	Link	2025-06-04	85.2	87.7	95.7	83.9	82.3	73.2	94.1	80.8	95.5	75.7	64.5	81.4	91.1
2	VL-Rethinker 🥈	Reason 🧠	Link	2025-04-10	80.3	-	-	-	-	-	-	-	-	-	-	-	-
3	Step R1-V-Mini 🥉	Reason 🧠	Link	2025-04-07	80.1	80.3	89.9	87.6	77.2	63.1	86.1	75.6	87.4	54.1	56.3	77.9	87.7
4	Kimi-k1.6-preview-20250308	Reason 🧠	Link	2025-03-10	80.0	82.9	91.8	82.3	76.0	63.1	87.2	73.1	89.1	45.9	52.8	77.0	90.7
5	Doubao-pro-1.5	Reason 🧠	Link	2025-01-22	79.5	77.7	88.9	86.0	82.3	62.0	89.7	73.5	88.3	35.1	55.3	77.1	86.5
6	Ovis2_34B	LMM 🖼️	Link	2025-02-10	77.1	75.5	84.6	89.2	72.8	62.0	80.1	72.2	82.0	37.8	54.9	73.8	85.0
7	Kimi-k1.5	Reason 🧠	Link	2025-01-22	74.9	-	-	-	-	-	-	-	-	-	-	-	-
8	OpenAI o1	Reason 🧠	Link	2024-09-12	73.9	-	-	-	-	-	-	-	-	-	-	-	-
9	Llama 4 Maverick	LMM 🖼️	Link	2025-04-05	73.7	-	-	-	-	-	-	-	-	-	-	-	-
10	Vision-R1-7B	Reason 🧠	Link	2025-03-09	73.2	75.1	82.7	80.1	70.3	54.7	78.6	65.2	79.1	40.5	45.1	68.9	83.1

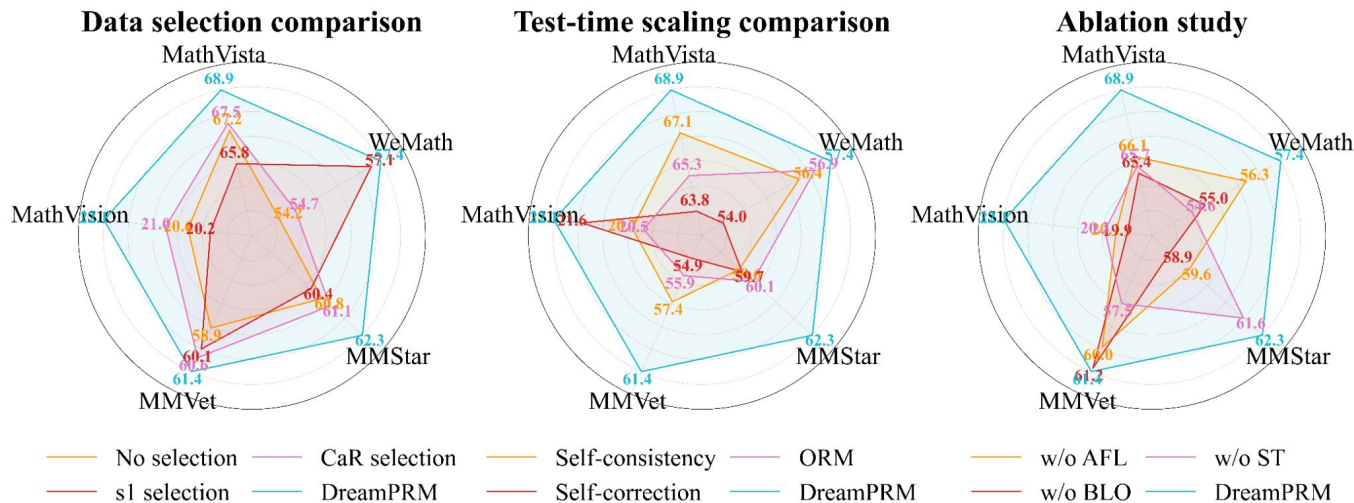
DreamPRM Performance with o4-mini

Method	MathVista accuracy
DreamPRM (ours) + o4-mini	85.2
Consensus + o4-mini (8 CoTs)	81.7
o4-mini (pass@1)	80.6

Results on Multi-modal Benchmarks

	Math Reasoning			General Reasoning	
	WEMATH (loose)	MATHVISTA (testmini)	MATHVISION (test)	MMVET (v1)	MMSTAR (test)
<i>Zero-shot Methods</i>					
Gemini-1.5-Pro [44]	46.0	63.9	19.2	<u>64.0</u>	59.1
GPT-4v [40]	51.4	49.9	<u>21.7</u>	67.7	<u>62.0</u>
LLaVA-OneVision-7B [23]	44.8	63.2	18.4	57.5	61.7
Qwen2-VL-7B [57]	42.9	58.2	16.3	62.0	60.7
InternVL-2.5-8B-MPO [58]	51.7	65.4	20.4	55.9	58.9
<i>Test-time Scaling Methods (InternVL-2.5-8B-MPO based)</i>					
Self-consistency [59]	56.4	67.1	20.7	57.4	59.6
Self-correction [16]	54.0	63.8	21.6	54.9	59.7
ORM [46]	56.9	65.3	20.5	55.9	60.1
Vanilla PRM [26]	54.2	67.2	20.6	58.9	60.8
CaR-PRM [15]	54.7	<u>67.5</u>	21.0	60.6	61.1
s1-PRM [38]	<u>57.1</u>	65.8	20.2	60.1	60.4
DreamPRM (ours)	57.4	68.9	22.1	61.4	62.3

Results on Multi-modal Benchmarks (2)



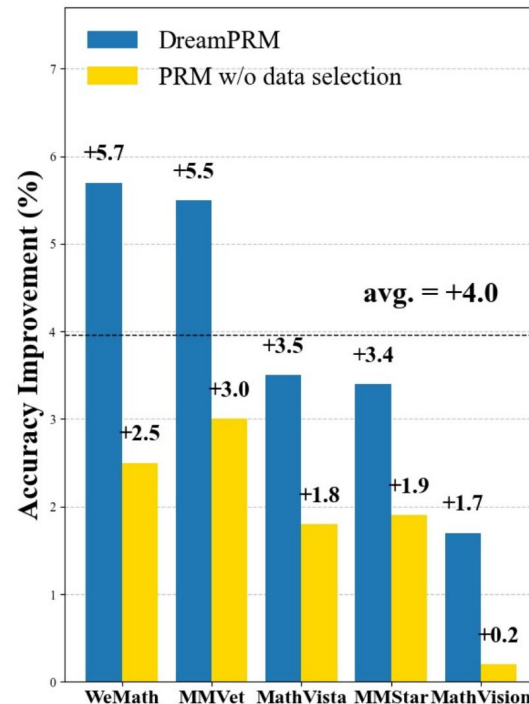
DreamPRM Results on Stronger MLLMs

Model Name	pass@1	DreamPRM (select k CoTs)				
	$k=1$	$k=2$	$k=4$	$k=6$	$k=8$	
InternVL-2.5-8B-MPO [58]	65.4	65.3	66.5	67.8	68.9	
GPT-4.1-mini (4-14-25) [40]	71.5	71.8	72.5	73.2	74.4	
o4-mini (4-16-25) [39]	80.6	81.6	82.5	84.2	85.2	

DreamPRM improves performance of cutting-edge LLMs,
including *GPT-4.1-mini* and *o4-mini*.

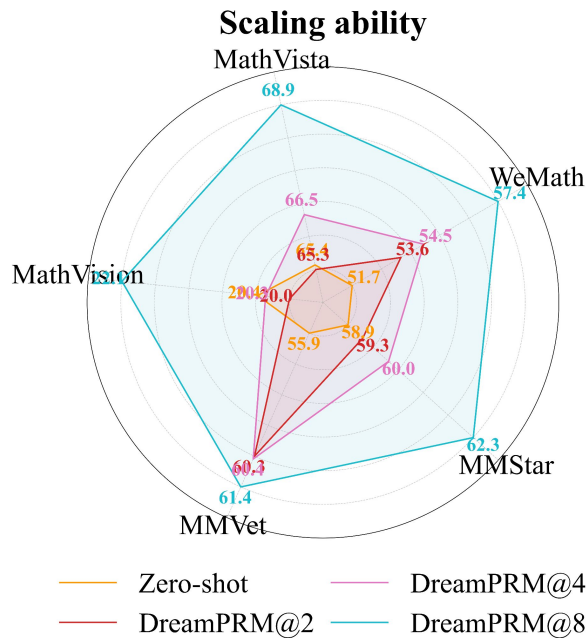
Effectiveness of Domain Reweighting

DreamPRM consistently surpasses Vanilla PRM trained without data selection.

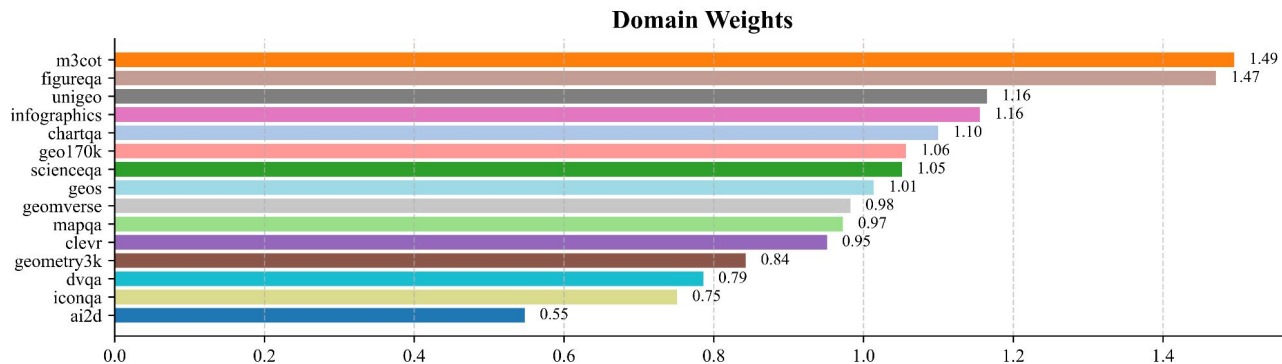


Scalability of DreamPRM

DreamPRM scales reliably with more CoT candidates.



Analysis of Learned Domain Weights



Challenging datasets like M3CoT and FigureQA receive high domain weights, while simple ones like AI2D receive low weights.

Thank you!
Any questions?